



ME 342A: SYSTEM DESIGN AND ANALYSIS

UNIT 1 SYSTEM FUNDAMENTAL CONCEPTS

Learning outcomes

- **System Definition, Examples**
- **System Approach**
- **Components of System**
- **Classification of Systems**
- **Characteristics of System**
- **System Analyst**
- **System Development Life Cycle**



System analysis is a process of fact gathering of the present system, diagnosing problems, analyzing the business requirements, and recommending improvements based on information gathered. System analysis creates a background and base for system design to achieve system objectives. Various components involved in system analysis are project initiation, preliminary investigation (fact gathering), feasibility analysis and input/ output report. Finally, system analysis brings out what a system should do.

After drafting out the complete requirements of a system in detail during the system analysis phase, system design focuses on how to accomplish the same to achieve system objectives. System design is a process of translating the user oriented functional specifications to technical design specifications or from logical design to physical design.

1 System Definition

The term “system” originates from the Greek term *systema*, which means to “place together”. A system is an integrated set of interoperable elements, each with explicitly specified and bounded capabilities, working synergistically to perform value-added processing to enable a user to satisfy mission-oriented operational needs in a prescribed operating environment with a specified outcome and probability of success.

By “an integrated set,” we mean that a system, by definition, is composed of hierarchical levels of physical elements, entities, or components.

By “interoperable elements,” we mean that elements within the system’s structure must be compatible with each other in *form*, *fit*, and *function*. System elements include equipment (e.g., hardware and software, personnel, facilities, operating constraints, support), maintenance, supplies, spares, training, resources, procedural data, external systems, and anything else that supports mission accomplishment.



By each element having “explicitly specified and bounded capabilities,” we mean that every element should work to accomplish some higher-level goal or purposeful mission. System element contributions to the overall system performance must be explicitly specified. This requires that *operational* and *functional* performance capabilities for each system element be identified and explicitly *bounded* to a level of specificity that allows the element to be analysed, designed, developed, tested, verified, and validated - either on a stand-alone basis or as part of the integrated system.

By “working in synergistically,” we mean that the purpose of integrating the set of elements is to *leverage* the capabilities of individual element capabilities to accomplish a higher-level capability that cannot be achieved as stand-alone elements.

By “value-added processing,” we mean that factors such operational cost, utility, suitability, availability, and efficiency demand that each system operation and task add *value* to its inputs availability, and produce outputs that contribute to achievement of the overall system mission outcome and performance objectives.

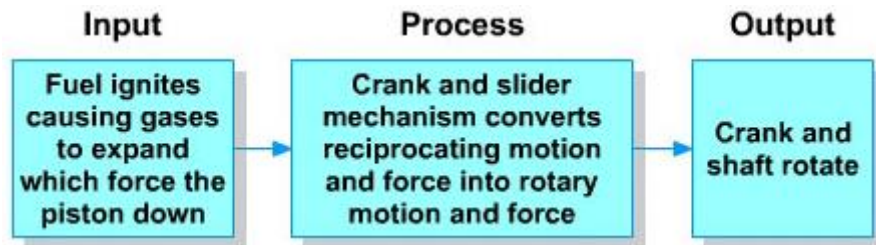
By “enable a user to predictably satisfy mission-oriented operational needs,” we mean that every system has a *purpose* (i.e., a reason for existence) and a *value* to the user(s). Its value may be a return on investment (ROI) relative to satisfying operational needs or to satisfy system missions and objectives.

Example: Mechanical system

A mechanical system is a set of physical components that convert an input motion and force into a desired output motion and force. Mechanical systems have at least three elements: input, process, and output.

- The input part of the system is any type of motion and force that drives the mechanical system. The input motion and force may be from any power source including human effort, energy from the wind, water, heat etc., from a chemical reaction or from an electrical, pneumatic, or hydraulic device.

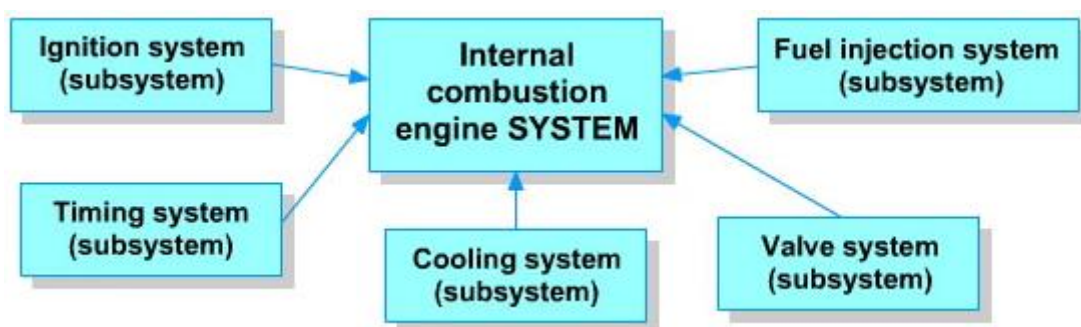
- The process part of the system is where mechanisms are used to convert the input motion and force into an output motion and force.
- The output is the change created in the input motion and force by the mechanism.



The system shown above is a small part of a larger and more complex system, e.g.

- For the fuel to ignite, an ignition system must be in place.
- The fuel must ignite at precisely the correct time so a timing system must also be in place.
- A measured amount of fuel must be injected into the combustion chamber at precisely the correct time so a fuel injection system must be in place.
- Exhaust gases must be evacuated from the combustion chamber so a valve system must be in place.

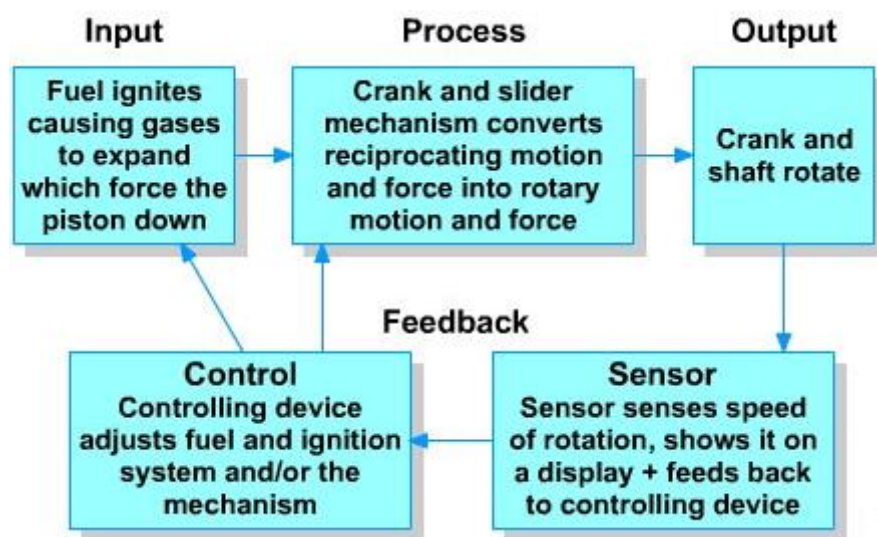
An internal combustion engine is a **system** that gives a motor vehicle the power to move. The ignition system, the timing system, the fuel injection system, and the valve system are **subsystems** of the internal combustion engine system. Subsystems are systems that are part of a larger system. Mechanical systems usually comprise of several subsystems.



Feedback and control

The system described above consists of input, process and output elements that result in a crankshaft rotating. The system does not allow for control over the speed of rotation of the crankshaft. To control the speed of rotation, there must be a monitoring and control subsystem built into the system. This can be achieved by:

- using sensors to monitor the input part of the system and feeding the information to a controlling device that makes changes to the input
- using sensors to monitor the mechanisms in the process part of the system and feeding the information to a controlling device that makes changes to the input and/or the process part of the system
- using sensors to monitor the output part of the system and feeding the information to a controlling device that makes changes to the input and/or the process part of the system.



2 Systems Approach

The systems approach is the foundation of systems analysis and design. It is fundamental to systems, systems thinking, systems methodology, systems design, and systems engineering. It is a set of top-level rules from which systems engineering methodologies can be derived.



The systems approach is a problem-solving paradigm. It considers the attributes of an entire system to achieve the objective of a system, which is to solve a problem. The systems approach allows the designer to manage, encapsulate, and anticipate complex behaviours. It allows the designer to anticipate and design for emergent behaviours.

The systems approach was introduced in the 1950s. Its features are used in psychology, sociology, political sciences, ecology, jurisprudence, systems thinking and systems engineering. The principles of the systems approach can be applied to all these fields. The systems approach pertains to both open and closed systems. The important aspect of open systems is that they interact with their environment and with other systems in their operational environment. Closed systems interact with the environment only slightly.

The following rules constitute the elements of the systems approach.

2.1 Identification of the elements of a system

A collection of interacting elements constitutes a system of interest (SOI). This SOI may interact with other sibling systems in an operations environment. When a system interacts with other systems, the combination of systems constitutes a system of systems. The elements of a system may consist of hardware, software, humans, processes, conceptual ideas, or any combination of these. The elements of a system may include systems beyond the operational system. Other elements may include, but are not limited to, the development system, the training system, the test system, the production system, and the support system.

2.2 Division of elements into smaller elements

The elaboration of elements into smaller elements allows the system to be viewed as a hierarchy. Although a system may be viewed in many ways, the hierarchical view is one of the more common. This hierarchical grouping reduces the perception of complication and complexity. Hierarchical grouping allows complexity to be contained, encapsulated, and concealed.



2.3 Grouping of elements

The grouping of elements within a system constitutes a sub-system. These sub-systems position themselves at a level appropriate to the problem being solved. Each grouping, or sub-system, is normally identified by the function this sub-system performs.

2.4 Identification of the boundary of a system

The boundary of a system is essential for defining the system itself. The boundary delineates the separation of the system of interest from its environment. It is the surface across which internal-to-external interactions take place. When the environment is a legal system or a social system, the boundary may be difficult to define. However, the definition of this boundary is the responsibility of the designer.

2.5 Identification of the function of each element

It is fundamental that every human-made system has a purpose and this purpose is reflected in the identification of the function of the system. It follows then that every element of the system also has a function. A system and its elements may have multiple functions and multiple elements may perform a single function jointly.

2.6 Identification of the interactions among the elements

The systems approach views the world as containing interacting systems and interacting subsystems. Analytic methodologies utilize interface analysis and management to incorporate this aspect of the systems approach. Consideration of interactions allows the designer to design in all the inflows in, the intra-flows, and the outflows from the system of interest.

2.7 Definition of the system's environment

The system environment is particularly important for open systems since an open system, by definition, exchanges energy, information, and material with its environment. Environments of interest include, but are not limited to, physical, cultural, economic, social, legal, political, and geographic environments. This



interaction occurs whether the system is being conceived, designed, synthesized, or operated. Definition of the system's environment allows the system designer to consider the dynamic context and situation and all exchanges, influences, and other factors.

2.8 Synthesis of the system

Synthesis is the construction of the entire system. Holistic methods are used to define the architecture of the entire system of interest. Synthesis allows the choice of alternatives at each level of the hierarchy. In the end, synthesis of the system of interest will result in maximum effectiveness of the system with a balanced solution satisfying the needs of all stakeholders. Synthesis utilizes iteration as a means of refinement and convergence. Synthesis leads to a robust and cost-effective solution.

The holistic view is basic to the systems approach. With the systems approach the starting point is the whole system. Holism is the philosophy that systems can best be understood. Holistic methods consist of those methods that consider multiple system elements and their interrelationships in the context of the whole. These methods are used, for example, to establish the basic architecture of the system. Holism also includes the evaluation of the environment on the entire system and on the consideration of the risk of the system.

2.9 Proving the system

Proving is an essential part of the systems approach. Proving establishes that the whole solution system performs with optimum effectiveness in its operational environment, while interacting with other systems in that environment, to solve the problem that prompted its creation. There are two parts to proving: verification and validation.



Verification is establishing that the system has been correctly created built to specification. Progressive integration and testing, using a simulated dynamic operational environment is a typical way to verify the system.

Validation determines whether the system meets the stakeholder needs. If possible, validation is performed in a real environment with real users and with a physical instance of the system. Validation emerges when the integrated system is tested as it operates and interacts with other systems in its (simulated) operational environment. Sometimes, as with some weapon systems, torpedoes, nuclear weapons, etc., it is not possible to validate the system in an actual environment, the simulated environment, containing other interacting systems, is used as a validation medium.

2.10 Identification of emergent characteristics of a system

Emergent characteristics are the properties of the whole that are not exclusively attributable to any of the interacting parts, and which are meaningless in the language appropriate at the level of those interacting parts. Emergent properties are identified by evaluating the interaction of the system with its environment. Emergent properties only exist at one level above the system level. It is possible to design in specific emergent properties, the various interacting subsystems, their interactions, and the environment in which the whole system will operate, including other systems with which it will interact in that environment need to be accounted for. Then, by adjusting the parts and their interactions at one level, emergence may be observed and measured at the next level.

So, creating requisite emergent properties means choosing the right parts, bringing them together in the right way, getting them to interact in the right way; and orchestrating those interactions.

3 Basic Components of a System



The basic components of a system are shown below.

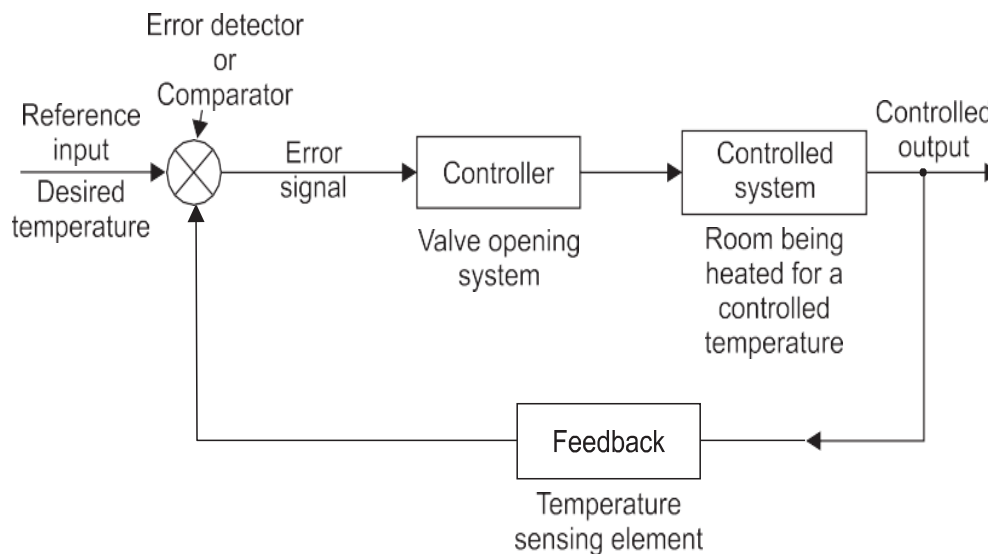


Fig. Representation of a system

Let us understand with an example of a room heating system using steam flowing through pipes fitted in the room. The flow of steam through the pipes is regulated automatically by a control valve. The amount of opening of the valve is regulated by a servomotor. The servomotor, controlling the opening or closing of the valve, works as a controller for amount of heating of the room. A temperature sensor will measure the room temperature and will provide a feedback signal for comparison in the comparator. In case the output temperature is different from the desired temperature a reference input is provided to the comparator, an error signal will be generated to control the hot steam flow through the valve.

The components shown in the diagram above are defined as follows:

1. Reference input: This provides input signal for the desired output.
2. Error detector: It is an element in which one system variable (feedback signal) is subtracted from another variable (reference signal) to obtain a third variable (error signal). It is also called comparator.
3. Feedback element: Feedback signal is a function of the controlled output which is compared with the reference signal to obtain the actuating signal.



Feedback element measures the controlled output, converts, or transforms to a suitable value for comparison with the reference input.

4. Error signal: It is an algebraic sum of the reference input and the primary feedback.
5. Controller: The controller is an element that is required to generate the appropriate control signal. The controller operates until the error between the controlled output and desired output is reduced to zero.
6. Controlled system: It is a body, a plant, a process, or a machine of which a condition is to be controlled, for example, a spacecraft, reactor, boiler, CNC machine, etc.
7. Controlled output: Controlled output is produced by the actuating signal available as input to the controller. Controlled output is made equal to the desired output with the help of the feedback system.

A major objective of a system is to produce an **output** that has value to its user. Whatever the nature of the output (goods, services, or information), it must be in line with the expectations of the intended user. Inputs are the elements (material, human resources, and information) that enter the system for processing. Output is the outcome of processing. A system feeds on input to produce output in much the same way that a business brings in human, financial, and material resources to produce goods and services. It is important to point out here that determining the output is a first step in specifying the nature, amount, and regularity of the input needed to operate a system. For example, in systems analysis, the first concern is to determine the user's requirements of a proposed computer system – that is, specification of the output that the computer is expected to provide for meeting user requirements.

The **processor** is the element of a system that involves the actual transformation of input into output. It is the operational component of a system. Processors may modify the input totally or partially, depending on the specifications of the output. This means that as the output specifications change



so does the processing. In some cases, input is also modified to enable the processor to handle the transformation.

The **control** element guides the system. It is the decision-making subsystem that controls the pattern of activities governing input, processing, and output. In an organizational context, management as a decision-making body controls the inflow, handling and outflow of activities that affect the welfare of the business. In a computer system, the operating system and accompanying software influence the behavior of the system. Output specifications determine what and how much input is needed to keep the system in balance. In systems analysis, knowing the attitudes of the individual who controls the area for which a computer is being considered can make a difference between the success and failure of the installation. Management support is required for securing control and supporting the objective of the proposed change.

Control in a dynamic system is achieved by **feedback**. Feedback measures output against a standard in some form of cybernetic procedure that includes communication and control. Output information is fed back to the input and/or to management (Controller) for deliberation. After the output is compared against performance standards, changes can result in the input or processing and consequently, the output. Feedback may be positive or negative, routing, or informational. Positive feedback reinforces the performance of the system. It is routine in nature. Negative feedback generally provides the controller with information for action. In systems analysis, feedback is important in different ways. During analysis, the user may be told that the problems in each application verify the initial concerns and justify the need for change. Another form of feedback comes after the system is implemented. The user informs the analyst about the performance of the new installation. This feedback often results in enhancements to meet the user's requirements.

The **environment** is the “supersystem” within which an organization operates. It is the source of external elements that impinge on the system. In fact,



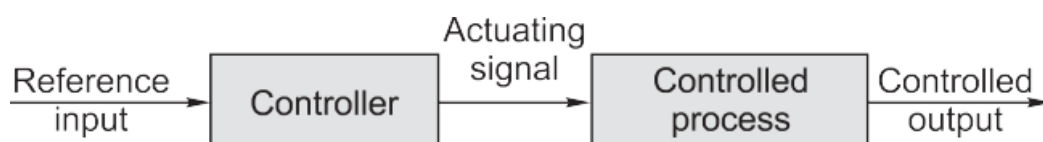
it often determines how a system must function. For example, the organization's environment, consisting of vendors, competitors, and others, may provide constraints and, consequently, influence the actual performance of the business.

A system should be defined by its **boundaries** – the limits that identify its components, processes, and interrelationship when it interfaces with another system. For example, a teller system in a commercial bank is restricted to the deposits, withdrawals and related activities of customers checking and savings accounts. It may exclude mortgage foreclosures, trust activities, and the like. Each system has boundaries that determine its sphere of influence and control. For example, in an integrated banking-wide computer system design, a customer who has a mortgage and a checking account with the same bank may write a check through the “teller system” to pay the premium that is later processed by the “mortgage loan system.” Recently, system design has been successful in allowing the automatic transfer of funds from a bank account to pay bills and other obligations to creditors, regardless of distance or location. This means that in systems analysis, knowledge of the boundaries of a given system is crucial in determining the nature of its interface with other systems for successful design.

4 Classification of Systems

4.1 Open-loop and closed-loop systems

The systems in which output has no effect on the control action are called open-loop systems. In other words, the output is neither measured nor fed back for comparison with reference input as shown in figure. For example, let us consider a washing machine in which soaking, washing, and rinsing in the washer



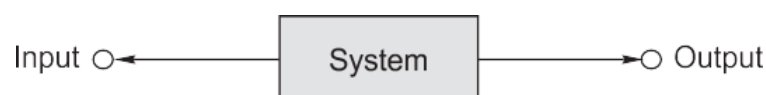
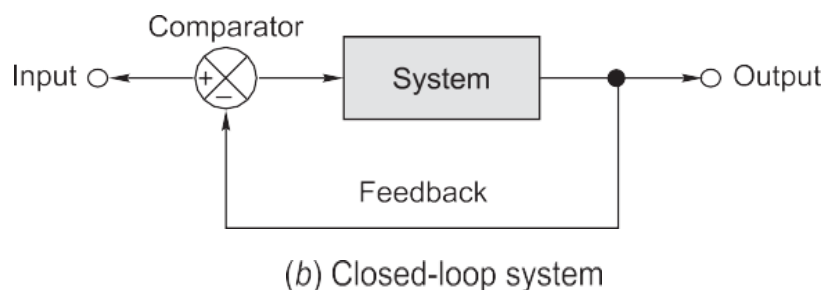


operates on the time basis. The machine does not measure the output signal, that is, cleanliness of clothes. Such open-loop systems can be used in practice if the relationship between input and output is known and there are no external or internal disturbances.

Fig. Elements of an open-loop system

Feedback control systems are referred to as closed-loop systems. In a closed-loop, the actuating error signal, which is the difference between the input signal and the feedback signal (output signal), is fed to the controller so as to reduce the error and bring the output of the system to the desired value. It is desirable that a closed-loop system is not influenced by external disturbances and internal variations in system parameters.

The operation of a system may be controlled externally or automatically (by the system itself). When the control action of a system is independent of the output, the system is said to be an *open-loop* system. However, if the control action is somehow dependent on the output, the system is called a *closed-loop* or *feedback* system. If the feedback signal is equal to the controlled output, the feedback system is called unity feedback system. Below figure shows an open-loop system and a closed-loop system. In a closed-loop system, the output is fed back and compared with the input so that any necessary corrective action may be taken. The open-loop system, however, relies for its action on the settings of its components.





(a) Open-loop system
Fig. Open-loop and closed-loop systems

Some examples of open-loop and closed-loop systems are as follows:

1. Traffic-light systems vary in operation. Some systems are closed-loop, and some are open-loop. In the open-loop type, there is a timing mechanism which is set to switch the lights at regular intervals, irrespective of the volume of traffic. In the closed-loop type, however, the amount of traffic passing through the junctions is monitored (electronically or otherwise), and the duration of the “red ” and “green” conditions are adjusted for each portion of the roads of the junction accordingly.
2. An example of the action of an open-loop system is a car-wash machine in which all cars receive the same amount of washing irrespective of how dirty they are. Here, the output is the cleanliness of the cars, which corresponds to a given setting of the machine (amount of water and washing time). A human being, however, who washes cars, automatically makes sure that the dirtier cars receive more attention than others. Car washing by a responsible human being is, therefore, a closed-loop system. An ordinary washing machine is an open-loop system.
3. A refrigerator is a closed-loop system. Its temperature is measured by a thermostat which turns the motor on when the temperature rises above the desired value and turns the motor off when the temperature again reaches the desired value.
4. The lighting of a room is an open-loop system. once the light is turned on, it will stay on (until it is switched off) irrespective of whether the room is dark or light.

The differences between an open-loop system and a closed-loop system are mentioned below in Table.

Table. Differences between Open-loop and Closed-loop Systems



Open-loop System	Closed-loop System
No feedback used	Feedback is there for comparison between desired output and reference input
Open-loop system is generally stable	Closed-loop system can become unstable under certain conditions
Their accuracy is determined by the calibration of their elements	They are more complex, complicated to construct and costly
Affected by non-linearities in the system	Adjusts to the effects of non-linearities present in the system
<i>Examples:</i> Washing machine, fixed time traffic control system, room heater	<i>Examples:</i> Servomotor control, generator output voltage control system, refrigerator, biological system

4.2 Linear and non-linear systems

A linear system is one which obeys the principle of superposition. The principle of superposition states that the response produced by simultaneous application of two different forcing functions is the sum of individual responses. Non-linear systems do not obey the principle of superposition. Almost all practical systems are non-linear to some extent. non-linearities are introduced due to saturation effect of system components, frictional forces, play between gear trains, mechanical linkages, non-linearities of electronic components like power amplifiers, transistors etc. used.

4.3 Time-invariant and time-varying systems

Time-invariant system is one whose parameters do not vary with time. The response of such a system is independent of the time at which input is applied. For example, resistance, inductance, and capacitance of an electrical network are independent of time.

A time-varying system is one in which one or more parameters vary with time. The response depends on the time at which input is applied. A space vehicle system where mass decreases with time, as fuel it carries is consumed during flight, is an example of a time-varying system.

4.4 Continuous time and discrete systems



In continuous systems, all system parameters are functions of continuous time, t . A discrete time control involves one or more variables that are known only at discrete instants of time. A continuous time or continuous data system is one in which the signals at various parts of the system are continuous functions of time. These signals are continuous time signals. For example, when we consider speed control of a DC motor, we know that the output, that is, the rotation in terms of radians per second is a function of voltage and current provided as input to it on a continuous time basis. Such a system takes a continuous time input and provides a continuous time output.

Sampled data system is one in which the signals at one or more points of the system are in the form of pulses or in digital code. Sampled data system will have discrete time input signal. Discrete data system is one in which a computer is used as one of the control circuit elements. The input and output are in binary numbers. Digital (A to D) converters also form a part of the system.

For example, in a generator excitation system, the field current is adjusted continuously based on feedback from the output voltage generated to keep the output voltage constant. Discrete signals are used in microprocessor and computer-based systems.

4.5 Single-Input–Single-Output (SISO) and Multi-Input–Multi-Output (MIMO) systems

A system with one input and one output is called single-input–single-output system. In other words, there is only one command and one controlled output. A system with multiple inputs and multiple outputs is called multi-input–multi-output system. For example, boiler drum level control, robot arm control, etc.

4.6 Lumped parameter and distributed parameter systems

Systems that can be described by ordinary differential equations are lumped parameter systems whereas distributed parameter systems are described by partial differential equations. The parameters of a long transmission line, that is, the resistance, inductance, and capacitance, are distributed along the line but



they may be considered as lumped parameters at certain points.

4.7 Deterministic and stochastic systems

A system is deterministic if the response is predictable and repeatable. If not, the system is a stochastic system which involves random variable parameters.

4.8 Static and dynamic systems

A system is called dynamic or time dependent if its present output depends on past input, whereas, a static system is the one whose current output depends only on current input. To use/understand a system efficiently, we must learn how to:

- (1) Develop mathematical system descriptions and reduce them to block diagram forms: (Modelling/Mathematical Representation).
- (2) Manipulate and solve the resulting system equations.
- (3) Design system to satisfy general performance specifications and
- (4) Evaluate our results by analytical and simulation studies.

It is, thus, important that we become familiar with a variety of electrical and mechanical systems.

5 Characteristics of a System

The definition of a system suggests some characteristics that are present in all systems: organization (order), interaction, interdependence, integration, and a central objective.

Organization implies structure and order. It is the arrangement of components that helps to achieve objectives. In the design of a business system, for example, the hierarchical relationships starting with the president on top and leading downward to the blue-collar workers represents the organization structure. Such an arrangement portrays a system-subsystem relationship, defines the authority structure, specifies the formal flow of communication, and formalizes the chain of command. Likewise, a computer system is designed around an input device, a central processing unit, an output device and one or



more storage units. When linked together they work as a whole system for producing information.

Interaction refers to the way each component functions with other components of the system. In an organization, for example, purchasing must interact with production, advertising with sales and payroll with personnel. In a computer system, the central processing unit must interact with the input device to solve a problem. In turn, the main memory holds programs and data that the arithmetic unit uses for computation. The interrelationship between these components enables the computer to perform.

Interdependence means that parts of the organization or computer system depend on one another. They are coordinated and linked together according to a plan. One subsystem depends on the input of another subsystem for proper functioning: that is, the output of one subsystem is the required input for another subsystem. This interdependence is crucial in systems work.

An integrated information system is designed to serve the needs of authorized users (department heads, managers, etc.) for quick access and retrieval via remote terminals. The interdependence between the personnel subsystem and the organization's users is obvious. No subsystem can function in isolation because it is dependent on the data (inputs) it receives from other subsystems to perform its required tasks. Interdependence is further illustrated by the activities and support of systems analysts, programmers, and the operations staff in a computer centre. A decision to computerize an application is initiated by the user, analyzed, and designed by the analyst, programmed, and tested by the programmer, and run by the computer operator. None of these persons can perform properly without the required input from others in the computer center subsystem.

Integration refers to the holism of systems. Synthesis follows analysis to achieve the central objective of the organization. Integration is concerned with how a system is tied together. It is more than sharing a physical part or location. It means that parts of the system work together within the system even though



each part performs a unique function. Successful integration will typically produce a synergistic effect and greater total impact than if each component works separately.

The last characteristic of a system is its **central objective**. Objectives may be real or stated. Although a stated objective may be the real objective, it is not uncommon for an organization to state one objective and operate to achieve another. The important point is that users must know the central objective of a computer application early in the analysis for a successful design and conversion. Political as well as organizational considerations often cloud the real objective. This means that the analyst must work around such obstacles to identify the real objective of the proposed change.

6 System Analyst

System analyst is a person who conducts a methodical study and evaluation of an activity such as a business to identify its desired objectives to determine procedures by which these objectives can be gained. The task of the system analyst is to elicit needs and resource constraints and to translate these into a viable operation. Designing and implementing systems to suit organizational needs are the functions of the system analyst. System analyst plays a vital role in analysis and design of a system or an application in any business/ scientific environment.

The analyst needs to identify problems by studying the existing system; collect all background information of the organization; study the historical background of the establishment; establish business and system objectives; gather all systems requirements and their specifications keeping in view the future expansion and other business requirements; find out various alternative feasible solutions and design suiting to organization resources; identify hardware and software requirements and their selection based on design recommended; design all input, output and procedures; develop the entire



system, test and implement it, and carry out the process till system becomes workable solution in accordance with the system objectives.

The system analyst is a person who is thoroughly aware of the system and guides the system development project by giving proper directions. He is an expert having technical and interpersonal skills to carry out development tasks required at each phase. He pursues to match the objectives of information system with the organization goal. The main roles of a system analyst are:

- a) Defining and understanding the requirement of user through various fact-finding techniques.
- b) Prioritizing the requirements by obtaining user consensus.
- c) Gathering the facts or information and acquires the opinions of users.
- d) Maintains analysis and evaluation to arrive at appropriate system which is more user friendly.
- e) Suggests many flexible alternative solutions, pick the best solution, and quantify cost and benefits.
- f) Draw certain specifications which are easily understood by users and programmer in precise and detailed form.
- g) Implemented the logical design of system which must be modular.
- h) Plan the periodicity for evaluation after it has been used for some time, and modify the system as needed.

Therefore, it is required that the system analyst must have several qualities and skill to play the role of investigator; change agent; psychology; architect; IT expert; trainer and motivator etc. To meet the challenge an analyst must have the following qualities and skills:

Interpersonal Skills

Interface with users and programmer.



Facilitate groups and lead smaller teams.

Managing expectations.

Good understanding, communication, selling and teaching abilities.

Motivator having the confidence to solve queries.

Analytical Skills

System study and organizational knowledge

Problem identification, problem analysis, and problem solving

Sound commonsense

Ability to access trade-off

Curiosity to learn about new organization

Management Skills

Understand users' jargon and practices.

Resource & project management.

Change & risk management.

Understand the management functions thoroughly.

Technical Skills

Knowledge of computers and software.

Keep abreast of modern development.

Know of system design tools.

Breadth knowledge about new technologies.

7 System Development Life Cycle method

The most prevalent and important approach for system analysis and design is known as System Development Life Cycle method (SDLC). Though it is traditional method but still it holds good in any methodology or approach as it is generic term for system development. This approach monitors and controls the system development beginning from fact gathering, designing, implementation and maintenance in cyclic way as a continuous process. SDLC enables users to transform a newly developed project into an operational one. It is a multistep,



iterative process structured in a methodical way. This process is used to model or provide a framework for technical and non-technical activities to deliver a quality system which meets or exceeds a business's expectations or manage decision-making progression.

An effective SDLC should result in a high-quality system that meets customer expectations, reaches completion within time and cost evaluations, and works effectively and efficiently in the current and planned Information Technology infrastructure. SDLC is a conceptual model which includes policies and procedures for developing or altering systems throughout their life cycles. SDLC is used by analysts to develop an information system. SDLC includes the following activities –

- requirements
- design
- implementation
- testing
- deployment
- operations
- maintenance

SDLC is a systematic approach which explicitly breaks down the work into phases that are required to implement either new or modified Information System.

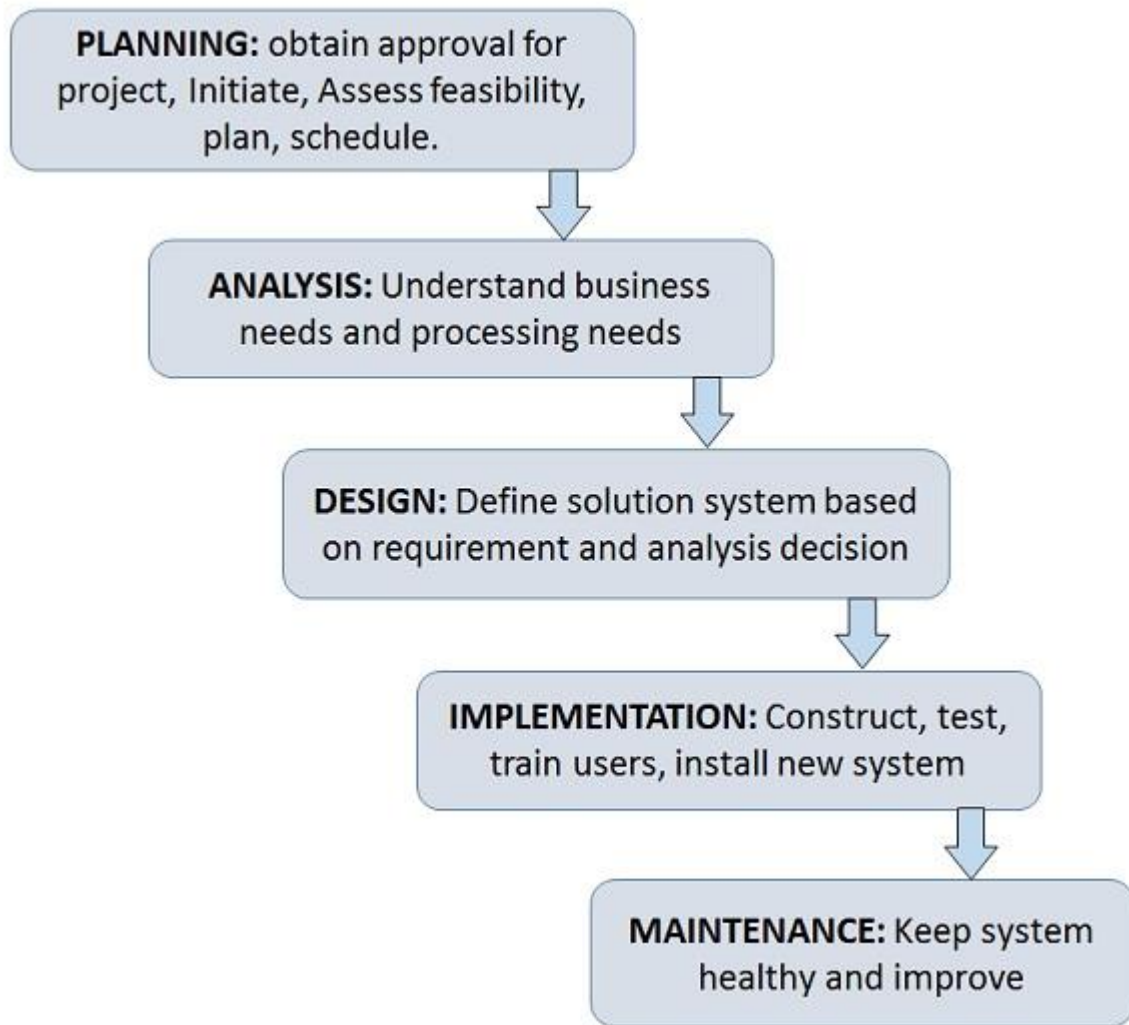


Fig. Phases of SDLC

Steps in various phases of SDLC

1. Feasibility Study or Planning

- Define the problem and scope of existing system.
- Overview the new system and determine its objectives.
- Confirm project feasibility and produce the project Schedule.
- During this phase, threats, constraints, integration, and security of system are also considered.
- A feasibility report for the entire project is created at the end of this phase.

2. Analysis and Specification

- Gather, analyse, and validate the information.



- b) Define the requirements and prototypes for new system.
- c) Evaluate the alternatives and prioritize the requirements.
- d) Examine the information needs of end-user and enhances the system goal.
- e) A Software Requirement Specification (SRS) document, which specifies the software, hardware, functional, and network requirements of the system is prepared at the end of this phase.

3. System Design

- a) Includes the design of application, network, databases, user interfaces, and system interfaces.
- b) Transform the SRS document into logical structure, which contains detailed and complete set of specifications that can be implemented in a programming language.
- c) Create a contingency, training, maintenance, and operation plan.
- d) Review the proposed design. Ensure that the final design must meet the requirements stated in SRS document.
- e) Finally, prepare a design document which will be used during next phases.

4. Implementation

- a) Implement the design into source code through coding.
- b) Combine all the modules together into training environment that detects errors and defects.
- c) A test report which contains errors is prepared through test plan that includes test related tasks such as test case generation, testing criteria, and resource allocation for testing.
- d) Integrate the information system into its environment and install the new system.

5. Maintenance/Support

- a) Include all the activities such as phone support or physical on-site support for users that is required once the system is installing.



- b) Implement the changes that software might undergo over a period of time or implement any new requirements after the software is deployed at the customer location.
- c) It also includes handling the residual errors and resolve any issues that may exist in the system even after the testing phase.
- d) Maintenance and support may be needed for a longer time for large systems and for a short time for smaller systems.

Review Questions

1. Define a system and represent it by a block diagram. Give examples of open-loop system and closed-loop system. State their differences.
2. Describe in details the various elements of a system with neat sketches and specify function of each element.
3. Classify the systems and categorize various systems we interact with in day to day life.
4. Represent in block diagram form, the system used in automatic control of speed of a steam turbine.
5. What is a feedback control system? Give two examples to show the use of systems in our everyday life.
6. State, giving reasons, whether the following are open-loop or closed-loop systems:
 - (1) The room heater or a hot-air blower used to heat a room in winter.
 - (2) The refrigerator.
 - (3) The air conditioning system used in electric trains.
 - (4) The traffic control system used in road crossings in typical Indian cities.



- (5) Maintenance of normal body temperature of a human being.
 - (6) Control of the speed of a motor vehicle by its driver depending upon road condition.
 - (7) The room air conditioner.
7. Human being is said to be the best example of an automatic control system. Explain.
8. Traffic at the road crossings are controlled using signal lamps (red, yellow, green). Represent through block diagram a closed-loop traffic flow control system.
9. Define the following terms: System, control system, feedback, closed-loop system, open- loop system, error detector, comparator, linear and non-linear systems, controller, and servomechanism.
10. The following are examples of open-loop systems:
- a. A room heater.
 - b. A traffic light control system.
 - c. A domestic washing machine.
 - d. A street light system.
 - e. A water pump lifting water to an overhead tank.

How will you change each of them into closed-loop systems?

- 11. Describe the System Life Cycle Diagram with suitable case study.
- 12. Highlight the role and attributes of a System Analyst.
- 13. Describe various phases of System Development Life Cycle.